# Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories, demographics, and spending patterns.

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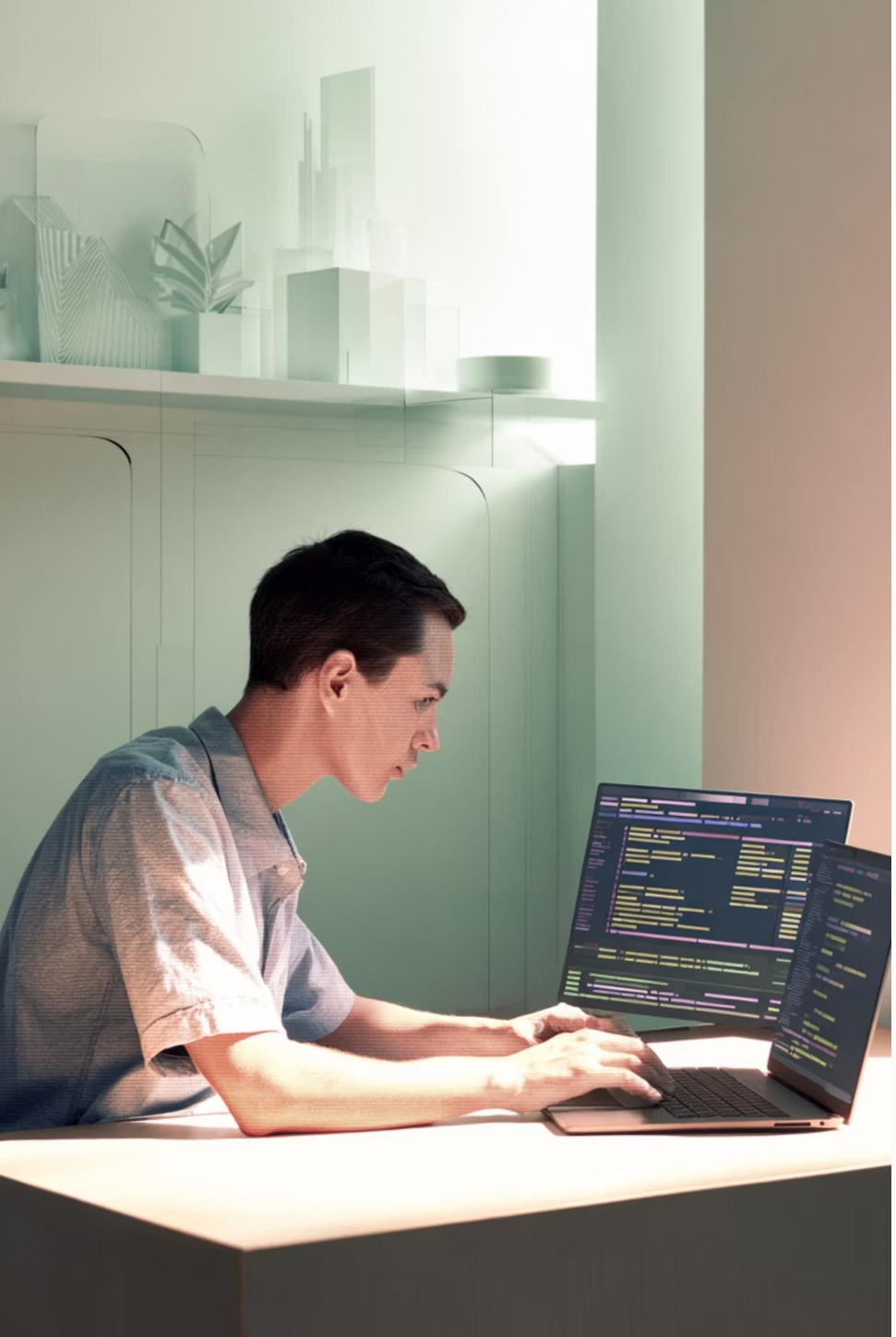
Customer Analytics Project
Python | SQL | Power BI

# Dataset at a Glance

| | |
|---|---|
| 3,900 | 18 |
| Transactions | Features |
| $59.76 | 3.75 |
| Avg Purchase | Avg Rating |

Key features: demographics, purchase details, shipping, discounts, subscriptions, and review ratings.



# Data Preparation in Python

01

## Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

## Clean & Impute

Filled 37 missing Review Ratings using category median

03

## Feature Engineering

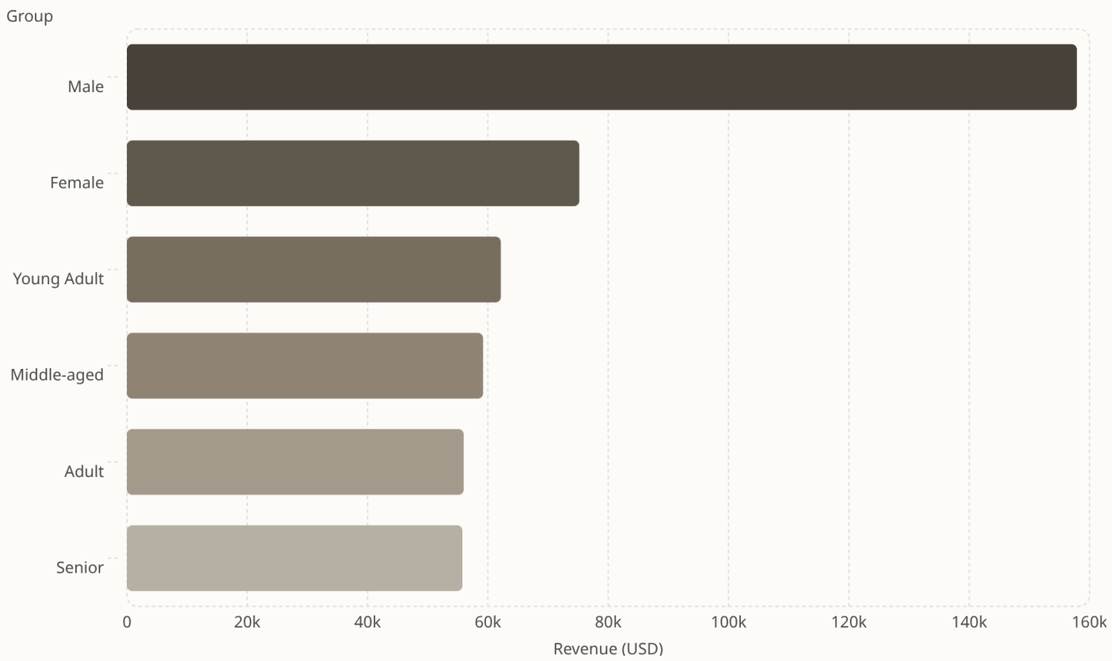
Created `age_group` and `purchase_frequency_days` columns

04

## Load to MySQL

Cleaned DataFrame pushed to database for SQL analysis

# Revenue Insights




## Key Findings

- Male customers generate **2× more revenue** than female
- Young Adults lead across all age groups
- Revenue is relatively balanced across age segments

# Shipping & Discount Behavior

## Shipping Type

**Standard**

Avg $58.46

**Express**

Avg $60.48

## Top Discount-Dependent Products

| Product | Discount Rate |
|---|---|
| Hat | 50.00% |
| Sneakers | 49.66% |
| Coat | 49.07% |
| Sweater | 48.17% |
| Pants | 47.37% |

# Top Products by Category & Rating

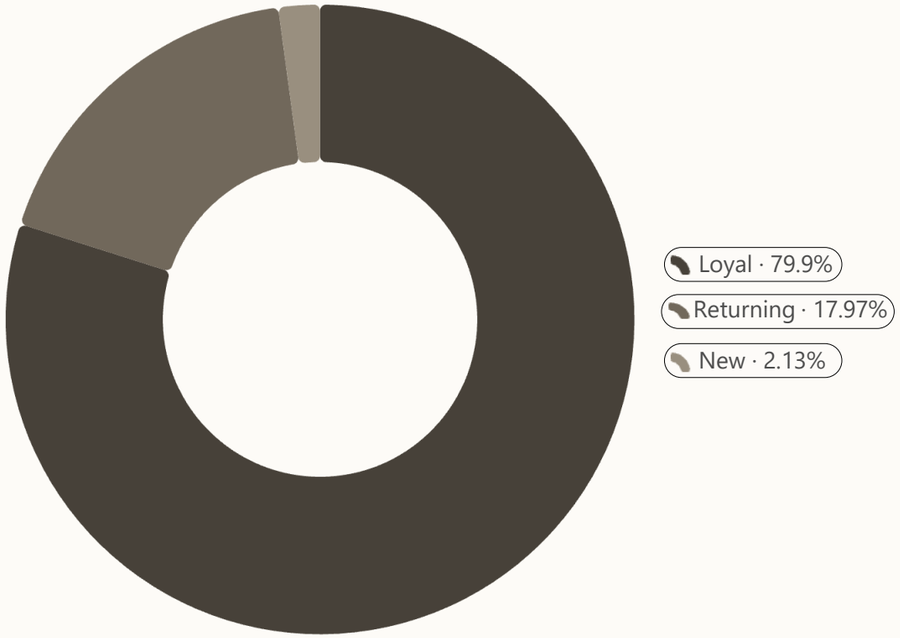
## Best-Rated Products

| Product | Avg Rating |
|---|---|
| Gloves | 3.86 |
| Sandals | 3.84 |
| Boots | 3.82 |
| Hat | 3.80 |
| Skirt | 3.78 |

## Top Sellers per Category

| Category | #1 Product |
|---|---|
| Accessories | Jewelry (171) |
| Clothing | Blouse (171) |
| Footwear | Sandals (160) |
| Outerwear | Jacket (163) |

# Customer Segmentation




## Segment Breakdown

### Loyal — 3,116

>5 previous purchases

### Returning — 701

Moderate purchase history

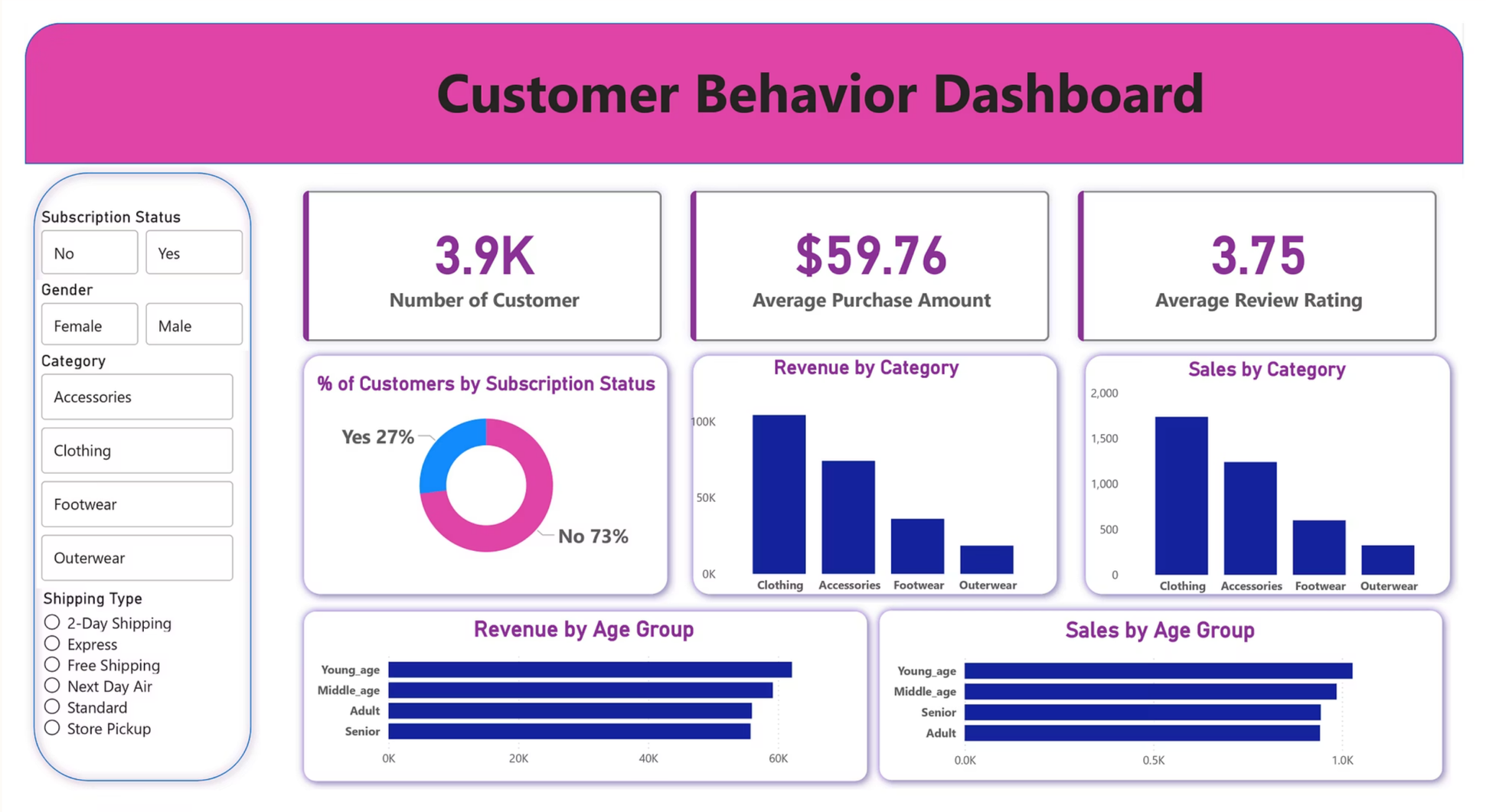
### New — 83

First-time buyers

# Subscription Analysis

| 27% | 73% | 958 |
|---|---|---|
| Subscribers | Non-Subscribers | Repeat Buyers Subscribed |
| 1,053 customers · Avg spend $59.49 | 2,847 customers · Avg spend $59.87 | vs. 2,518 repeat buyers not subscribed |

Repeat buyers (>5 purchases) are **undersubscribed** — a key opportunity for conversion campaigns.




# Power BI Dashboard

Interactive dashboard with filters by subscription, gender, category, and shipping type — visualizing KPIs, revenue by category, and sales by age group.

# Business Recommendations

### Boost Subscriptions

Promote exclusive benefits to convert repeat buyers

### Loyalty Programs

Reward returning customers to grow the Loyal segment

### Review Discount Policy

Balance sales boosts with margin control

### Targeted Marketing

Focus on high-revenue age groups and express-shipping users

### Product Positioning

Highlight top-rated and best-selling items in campaigns